

Fundamentals of API Development (4 Sessions × 4 Hours, Delivered Across 4 Days, AM/PM)

Duration: 4 Sessions × 4 Hours (16 hours total, split across 4 days, AM and PM cohorts)

Audience: Software developers who need the fundamentals of API design and development

Track: Java (Spring Boot). Hands-on labs are built against a Spring Boot REST API, tested with JUnit5 and Postman, using IntelliJ as the IDE.

Prerequisites:

- Basic Java familiarity (classes, interfaces, collections)
- IntelliJ, a JDK, and Postman installed before class
- No prior API design experience required



Delivery Model: 1 Course, 2 Cohorts, 4 Days

This is authored as a single 16-hour course, but delivered as four 4-hour sessions on four separate days so two cohorts (morning and afternoon) can each go through the full course without doubling instructor prep:

	Day 1	Day 2	Day 3	Day 4
Morning Cohort	Session 1 (AM)	Session 2 (AM)	Session 3 (AM)	Session 4 (AM)
Afternoon Cohort	Session 1 (PM, repeated)	Session 2 (PM, repeated)	Session 3 (PM, repeated)	Session 4 (PM, repeated)

Each cohort attends one slot per day (AM or PM) and follows Session 1 through Session 4 across the four days, a 4-day course for that cohort. The instructor delivers the same session content twice each day, once for the AM cohort and once for the PM cohort.



Course Overview

This course teaches the fundamentals of API design and development: what APIs are and why they matter, how REST/SOAP/RPC protocols differ, how to write and document API contracts, how to manage an API's lifecycle and secure it, and how APIs fit into a modern DevOps ecosystem. Hands-on labs use a Spring Boot REST API as the shared example, with Postman for exploring/testing endpoints, JUnit5 for automated tests, and IntelliJ as the development environment throughout.



Main Topics Covered

- APIs: Purpose, Nomenclature & Types
 - Benefits and drawbacks of APIs, key terms, API types (Open, Partner, Internal)

- API Protocols & RESTful Basics
 - REST, SOAP, RPC overview; standards (HTTP methods, status codes, resource naming)
- API Contracts & Documentation
 - OpenAPI/Swagger standards; documenting APIs and common tooling
- API Lifecycle & Versioning
 - Common lifecycle approaches, pros/cons, non-breaking version strategies
- Security for APIs
 - Authentication, authorization, rate limiting
- DevOps & the API Ecosystem
 - API First strategy, CI/CD, automated testing, security within the ecosystem



Learning Outcomes

- Explain the purpose of APIs and define key terms in API development
- Differentiate between API types and API protocols
- Build contracts into APIs and describe RESTful standards
- Discuss the API lifecycle and design a non-breaking versioning strategy
- Understand why security matters for APIs and configure basic security rules
- Describe how APIs fit into a DevOps/CI/CD ecosystem



Project Context

Labs build on a shared Spring Boot REST API example, extended lab by lab. You'll work in IntelliJ as the IDE, explore and test endpoints in Postman, write automated tests in JUnit5, and use a Swagger/OpenAPI editor for the contract and documentation labs.



Day 1 (4 Hours): API Purpose, Types & Protocols

Delivered once for the morning cohort and once (repeated) for the afternoon cohort.

1. Welcome, Learner Introductions & Course Objectives (15 min)

Course arc preview: what's covered across the 4 days.

2. APIs: Purpose, Nomenclature & Types (35 min)

- APIs: purpose and nomenclature
- Benefits and drawbacks of APIs
- Key terms in API development
- API types: Open, Partner, Internal

Kahoot 1 (10 min)

Bio Break (10 min)

3. API Protocols Overview: REST, SOAP & RPC (30 min)

Bio Break (10 min)

4. RESTful APIs: The Basics (35 min)

- Drawbacks of not having standards
- Common standards: HTTP methods, status codes, resource naming

5. Breakout 1: Lab 1.1: Analyzing an Existing API (30 min)

- Analyze an existing API to identify its protocol and standards adherence
- Stretch goals: compare a second API against the same checklist; document where it deviates from REST conventions

Bio Break (10 min)

Kahoot 2 (10 min)

6. Discussion & Buffer: API Types/Protocols Deep-Dive, Q&A (35 min)

- Open discussion on API types and protocol trade-offs from Lab 1.1
- Flex time: catch-up room for cohorts running behind, or extended Lab 1.1 stretch goals

7. Day 1 Wrap-up & Day 2 Preview (10 min)

Day 1 Deliverable: Working understanding of API purpose, types, and protocols; a completed protocol/standards analysis of an existing API (Lab 1.1).



Day 2 (4 Hours): API Contracts, Documentation & Design

Delivered once for the morning cohort and once (repeated) for the afternoon cohort. Continuation of Day 1 for the same cohort.

1. Day 2 Kickoff & Day 1 Recap (10 min)

2. API Contracts: Standards & OpenAPI/Swagger (35 min)

- Drawbacks of not having standards
- Summary of common standards (OpenAPI/Swagger)

Kahoot 1 (10 min)

Bio Break (10 min)

3. Breakout 1: Lab 1.2: Writing an OpenAPI Specification (30 min)

- Write an OpenAPI specification for a basic resource
- Stretch goals: add request/response examples and error schemas

Bio Break (10 min)

4. Documenting APIs: Standards & Tools (35 min)

- Drawbacks of not having standards
- Summary of common standards and tools

5. Breakout 2: Lab 1.3: Peer Review & Interactive Docs (30 min)

- Peer review of API contracts
- Generate interactive documentation (Swagger UI)
- Stretch goals: annotate edge cases; add auth documentation to the spec

Bio Break (10 min)

Kahoot 2 (10 min)

6. Design Lab Discussion: Contract-First vs. Code-First (40 min)

- Comparing contract-first and code-first workflows using Postman/Swagger tooling
- Group review of Lab 1.2/1.3 outputs

7. Day 2 Wrap-up & Day 3 Preview (10 min)

Day 2 Deliverable: An OpenAPI specification for a basic resource, peer-reviewed, with generated interactive documentation (Labs 1.2-1.3).



Day 3 (4 Hours): API Lifecycle & Security

Delivered once for the morning cohort and once (repeated) for the afternoon cohort. Continuation of Day 2 for the same cohort.

1. Day 3 Kickoff & Day 2 Recap (10 min)

2. API Lifecycle: Overview & Versioning (35 min)

- API lifecycle overview
- API versioning
- Review of most common approaches to lifecycle (pros and cons)

Kahoot 1 (10 min)

Bio Break (10 min)

3. Breakout 1: Lab 2.1: Designing a Non-Breaking Version Strategy (30 min)

- Design a non-breaking version strategy for an evolving API
- Stretch goals: draft a deprecation/sunset policy; handle a breaking-change edge case

Bio Break (10 min)

4. Security for APIs: Standards (35 min)

- Drawbacks of not having standards

- Common standards: authentication, authorization, rate limiting

5. Breakout 2: Lab 2.2: Implementing Basic API Security Rules (30 min)

- Implement or configure basic API security rules on the Spring Boot API
- Stretch goals: add rate-limit tiers; validate auth flows with a Postman collection

Bio Break (10 min)

Kahoot 2 (10 min)

6. Lifecycle & Security Standards Deep-Dive, Q&A (40 min)

- Drawbacks of skipping standards for both lifecycle and security, with real-world examples
- Group review of Lab 2.1/2.2 outputs

7. Day 3 Wrap-up & Day 4 Preview (10 min)

Day 3 Deliverable: A non-breaking versioning strategy (Lab 2.1) and configured basic security rules (Lab 2.2) on the shared API.



Day 4 (4 Hours): DevOps, API Ecosystem & Capstone

Delivered once for the morning cohort and once (repeated) for the afternoon cohort. Continuation of Day 3 for the same cohort.

1. Day 4 Kickoff & Day 3 Recap (10 min)

2. DevOps in API Development: Overview (30 min)

Kahoot 1 (10 min)

Bio Break (10 min)

3. API Ecosystem (35 min)

- API First strategy
- DevOps (CI/CD, automated testing)
- API security within the ecosystem
- The case for deep knowledge

Bio Break (10 min)

4. Breakout 1: Lab 2.3: Simulating an Automated API Test Suite (30 min)

- Build a JUnit5 test suite for the Spring Boot API, run from IntelliJ
- Stretch goals: add negative test cases; integrate a Postman collection runner

Kahoot 2 (10 min)

Bio Break (10 min)

5. Breakout 2: Lab 2.4: Capstone Exercise (40 min)

Capstone exercise integrating design, documentation, versioning, security, and testing from Days 1-4.

6. Review of Capstone Solutions & Discussion (35 min)

7. Course Recap, Final Q&A & Next Steps (10 min)

Day 4 Deliverable: Completed capstone integrating contract design, documentation, versioning, security, and an automated JUnit5 test suite.

Session Breakdown Table

Day 1 (4 hours / 240 min)

Topic	Duration
Welcome, Learner Introductions & Course Objectives	15 min
APIs: Purpose, Nomenclature & Types	35 min
Kahoot 1	10 min
Bio Break	10 min
API Protocols Overview: REST, SOAP & RPC	30 min
Bio Break	10 min
RESTful APIs: The Basics	35 min
Breakout 1: Lab 1.1: Analyzing an Existing API	30 min
Bio Break	10 min
Kahoot 2	10 min
Discussion & Buffer: API Types/Protocols Deep-Dive, Q&A	35 min
Day 1 Wrap-up & Day 2 Preview	10 min
Total	240 min

Day 2 (4 hours / 240 min)

Topic	Duration
Day 2 Kickoff & Day 1 Recap	10 min
API Contracts: Standards & OpenAPI/Swagger	35 min
Kahoot 1	10 min

Topic	Duration
Bio Break	10 min
Breakout 1: Lab 1.2: Writing an OpenAPI Specification	30 min
Bio Break	10 min
Documenting APIs: Standards & Tools	35 min
Breakout 2: Lab 1.3: Peer Review & Interactive Docs	30 min
Bio Break	10 min
Kahoot 2	10 min
Design Lab Discussion: Contract-First vs. Code-First	40 min
Day 2 Wrap-up & Day 3 Preview	10 min
Total	240 min

Day 3 (4 hours / 240 min)

Topic	Duration
Day 3 Kickoff & Day 2 Recap	10 min
API Lifecycle: Overview & Versioning	35 min
Kahoot 1	10 min
Bio Break	10 min
Breakout 1: Lab 2.1: Designing a Non-Breaking Version Strategy	30 min
Bio Break	10 min
Security for APIs: Standards	35 min
Breakout 2: Lab 2.2: Implementing Basic API Security Rules	30 min
Bio Break	10 min
Kahoot 2	10 min
Lifecycle & Security Standards Deep-Dive, Q&A	40 min
Day 3 Wrap-up & Day 4 Preview	10 min
Total	240 min

Day 4 (4 hours / 240 min)

Topic	Duration
Day 4 Kickoff & Day 3 Recap	10 min
DevOps in API Development: Overview	30 min
Kahoot 1	10 min
Bio Break	10 min
API Ecosystem	35 min
Bio Break	10 min
Breakout 1: Lab 2.3: Simulating an Automated API Test Suite	30 min
Kahoot 2	10 min
Bio Break	10 min
Breakout 2: Lab 2.4: Capstone Exercise	40 min
Review of Capstone Solutions & Discussion	35 min
Course Recap, Final Q&A & Next Steps	10 min
Total	240 min

Combined Total Duration: 16 hours (960 minutes) across 4 sessions / 4 days, run twice per day for 2 cohorts.

Deliverables (End of Course)

- Protocol/standards analysis of an existing API (Lab 1.1)
- OpenAPI specification for a basic resource, peer-reviewed, with interactive docs (Labs 1.2-1.3)
- Non-breaking versioning strategy (Lab 2.1)
- Configured basic API security rules (Lab 2.2)
- JUnit5 automated test suite for the shared Spring Boot API (Lab 2.3)
- Capstone integrating design, documentation, versioning, security, and testing (Lab 2.4)

Teaching Philosophy

The course uses one Spring Boot API example throughout, so each lab builds on the last, tested with Postman and JUnit5 in IntelliJ.